THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

SEPM Assignment 1

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Roll - 48 Batch - T13

Q) Explain Software process models

→ 1) Waterfall Model Software Engineering

Requirements
↳ Design
↳ Development
↳ Testing
↳ Deployment
↳ Maintainance

i) Requirement: Customer requirement are gathered analyzed to remove errors and Conflicts, and documented in the SRS, which acts as a Contract between Customer and developers

ii) System Design: Requirements are Converted into a design that Can be Coded includes High-Level Design (HLD) for overall architecture and low level Design (LLD) for detailed Component design, documented in the SDD

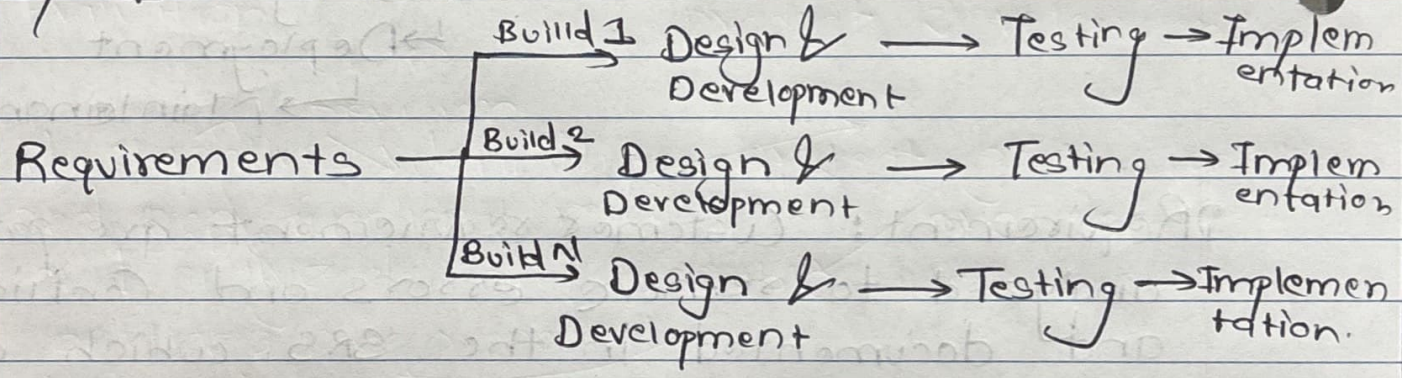
iii) Development: Designs are translated into Source code using suitable programming languages. Each module is coded and unit tested to ensure Correct functionality.

iv) Testing: Complete System testing is performed (Alpha, Beta, Acceptance testing

v) Deployment : After successful testing, the Software is deployed to users with necessary setup and training

vi) Maintenance : Post deployment changes are handled, including Corrective, perfective, and Adaptive maintenance to fix bugs, enhance features and adapt to new environment

2) Incremental Model



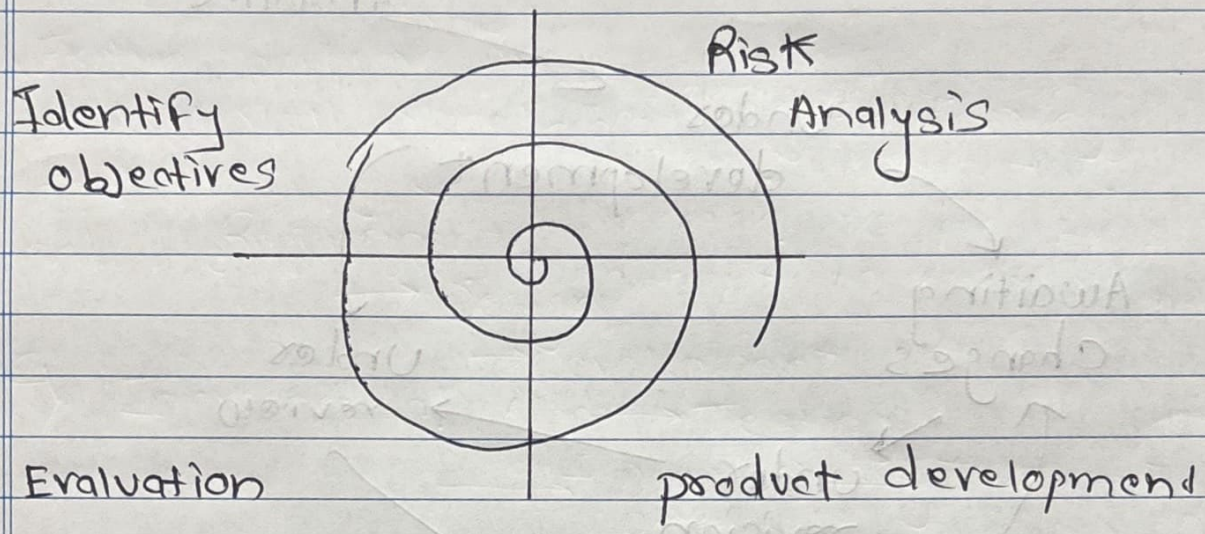
i) Requirement : Requirements are gathered from product experts and analysed to clearly define the systems functional needs, forming the foundation for development.

ii) Design & Development : The system design is prepared and new features are developed incrementally, ensuring the software functions as expected.

iii) Testing : Each developed feature is tested using appropriate testing methods to verify both new and existing functionalities.

iv) Implementation : Final Code is written and verified. The software is gradually improved through increments untill the complete working product is achieved.

3) Spiral Model



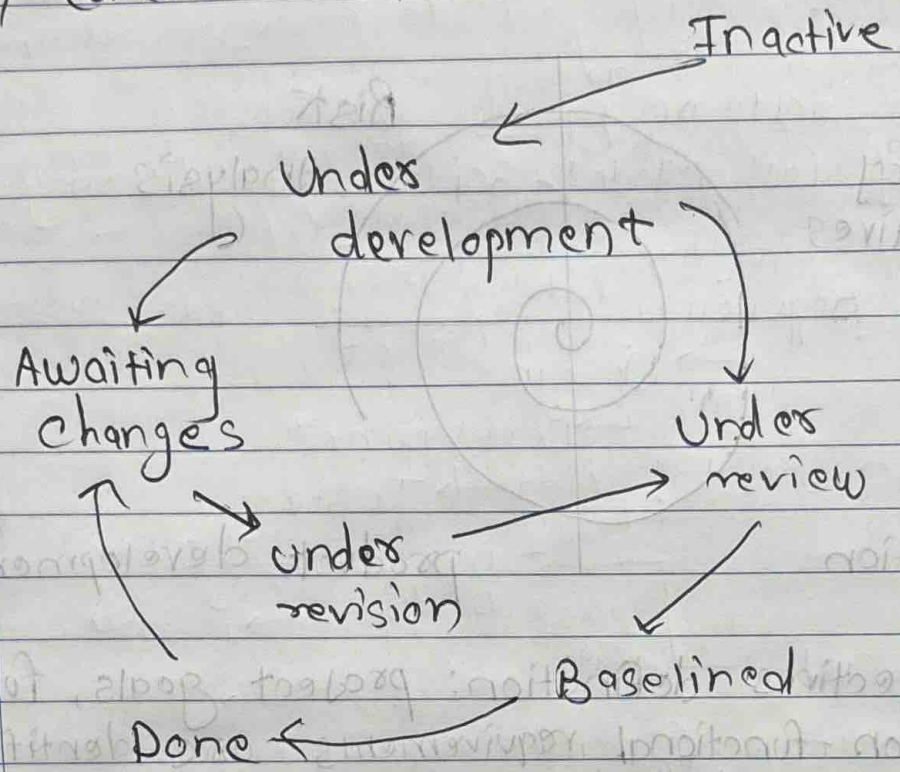
i) objectives defination: project goals, functional and non-functional requirements are identified and analyzed, and possible alternative solutions are proposed

ii) Risk Analysis: Potential risks are identified and evaluated for each solution. The best solution is selected, risks are resolved, and a prototype is built.

iii) Development & Testing: Identified features are developed and tested. A working version of the software is produce at the end

iv) Evaluation: The developed software is evaluated by the customer to check quality and requirement fulfilment.

4) Concurrent Model:



This model allows multiple phases at development to occurs simultaneously

Working principle:
- Each activity exists in different states.
- Activities progress in dependently but are Co-ordinated

Adv - Faster development
suitable for parallel work

Dis Adv: Complex to manage
Requires effective Communication